FROM COMPLEX ANALYSIS TO THE GROUP STRUCTURE ON ELLIPTIC CURVE

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Abstract. This note aims to derive the group structure on elliptic curves using elementary tools such as Liouville's theorem and argument principle.

1. Lattice and Elliptic functions

Let $w_1, w_2 \in \mathbb{C}$ such that $\Im(w_1/w_2) > 0$. We define the *lattice* associated with w_1, w_2 as $L \coloneqq w_1\mathbb{Z} \oplus w_2\mathbb{Z} \subset \mathbb{C}$.

Associated to this lattice L, we define a *complex torus* as the quotient space $\mathbb{C}/L$. This space inherits the natural structure of a Riemann surface. Within this framework, a meromorphic function on the complex torus can be naturally identified with a meromorphic function on $\mathbb{C}$ possessing a double period. These are termed elliptic functions.

Specifically, an elliptic function associated with lattice L is defined as a meromorphic function f on $\mathbb{C}$ satisfying $f(x) = f(y)$ for any $x, y \in \mathbb{C}$ such that $x - y \in L$. We denote the set of all such functions by $\mathcal{E}_L$.

Remark 1.1. The set $\mathcal{E}_L$ forms a field closed under differentiation.

We introduce some notations initially. Let Π represent the parallelogram in $\mathbb{C}$, defined as $\{(sw_1, tw_2) : s, t \in [0, 1]\}$. For a given $\alpha \in \mathbb{C}$, consider a function f from the set $\mathcal{E}_L$, which exhibits neither zeros nor poles on the boundary $\partial(\alpha+\Pi)$, denoted as $\alpha+\Pi = \{\alpha+z : z \in \Pi\}$. The existence of such α is guaranteed, as zeros and poles in Π are discrete and thus finite, given Π's compactness.

Lemma 1.2. *For any $f \in \mathcal{E}_L$ that is holomorphic on $\partial(\alpha + \Pi)$, we have $\int_{\partial(\alpha+\Pi)} f = 0$.*

Proof. The integral $\int_{\partial(\alpha+\Pi)} f$ can be decomposed into the sum of integrals over each side of the parallelogram:

$$\int_{\alpha}^{\alpha+w_1} f + \int_{\alpha+w_1}^{\alpha+w_1+w_2} f + \int_{\alpha+w_1+w_2}^{\alpha+w_2} f + \int_{\alpha+w_2}^{\alpha} f.$$

Since f is elliptic, by change of variable

$$\int_{\alpha+w_1}^{\alpha+w_1+w_2} f(z)dz = \int_{\alpha}^{\alpha+w_2} f(z + w_1)d(z + w_1) = \int_{\alpha}^{\alpha+w_2} f(z)dz.$$

Likewise,

$$\int_{\alpha+w_2}^{\alpha+w_1+w_2} f(z)dz = \int_{\alpha}^{\alpha+w_1} f(z)dz.$$

This implies $\int_{\partial(\alpha+\Pi)} f = 0$. $\square$

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Proposition 1.3. *For any elliptic function $f \in \mathcal{E}_L$, there are equal numbers of zeros and poles, counted with multiplicities, within the parallelogram Π.*

Proof. It suffices to verifies this in $\alpha + \Pi$, since f is elliptic. Given f has neither zeros nor poles on $\partial(\alpha + \Pi)$, the Argument Principle implies

$$\#Z(f) - \#P(f) = \int_{\partial(\alpha+\Pi)} \frac{f'}{f}$$

, where $\#Z(f), \#P(f)$ denote zeros and poles of f up to multiplicities in Π respectively. Notice $\frac{f'}{f} \in \mathcal{E}_L$ thus it follows that

$$\#Z(f) - \#P(f) = \int_{\partial(\alpha+\Pi)} \frac{f'}{f} = 0$$

by Lemma1.2. □

Proposition 1.4. *For every elliptic function $f \in \mathcal{E}_L$, the sum $\sum_{z\in\alpha+\Pi} ord_z(f) \cdot z$ is an element of the lattice L, where $ord_z(f)$ represents the order of f at the point z.*

Proof. By the generalized argument principle, we have

$$\sum_{z\in\alpha+\Pi} \mathrm{ord}_z(f) \cdot z = \frac{1}{2\pi i} \int_{\partial(\alpha+\Pi)} \frac{f'}{f} z dz.$$

We select α such that f does not vanish on the boundary. For integrations, consider

$$\int_{\alpha+w_1}^{\alpha+w_1+w_2} \frac{f'}{f} z dz = \int_{\alpha}^{\alpha+w_2} \frac{f'}{f}(t + w_1) dt = \int_{\alpha}^{\alpha+w_2} \frac{f'}{f} t dt + w_1 \int_{\alpha}^{\alpha+w_2} \frac{f'}{f} t dt.$$

Similarly,

$$\int_{\alpha+w_2}^{\alpha+w_1+w_2} \frac{f'}{f} z dz = \int_{\alpha}^{\alpha+w_1} \frac{f'}{f} t dt + w_2 \int_{\alpha}^{\alpha+w_1} \frac{f'}{f} t dt.$$

This implies

$$\sum_{z\in\alpha+\Pi} \mathrm{ord}_z(f) \cdot z = \frac{1}{2\pi i} \int_{\partial(\alpha+\Pi)} \frac{f'}{f} z dz = w_1 \int_{\alpha}^{\alpha+w_2} \frac{f'}{f} t dt + w_2 \int_{\alpha}^{\alpha+w_1} \frac{f'}{f} t dt.$$

On the other hand,

$$\frac{1}{2\pi i} \int_{\alpha}^{\alpha+w_1} \frac{f'}{f} dz = \frac{1}{2\pi i} \int_{f\gamma} \frac{dz}{z} \in \mathbb{Z}$$

and similarly

$$\frac{1}{2\pi i} \int_{\alpha}^{\alpha+w_2} \frac{f'}{f} dz \in \mathbb{Z}.$$

Consequently,

$$\sum_{z\in\alpha+\Pi} \mathrm{ord}_z(f) \cdot z \in L.$$

□

Remark 1.5. Proposition 1.4 will be interpreted as a weighted average of zeros and poles of a given elliptic function.

2. Example of elliptic function: Weierstrass p-function

Define Weierstrass function

$$\wp(z) = \frac{1}{z^2} + \sum_{w \in L \setminus \{0\}} \left(\frac{1}{(w-z)^2} - \frac{1}{w^2}\right).$$

It is evidently even. To see it is elliptic, consider

$$\wp'(z) = \sum_{w \in L} \frac{-2}{(z-w)^3},$$

which is elliptic. That is Therefore,

$$\wp(z + w_i) - \wp(z) = constant, i = 1, 2.$$

This constant is 0 since $\wp$ is even. To sum up, $\wp \in \mathcal{E}_L$.

Remark 2.1.
This function is convergent; we will leave the details to readers. See also [2][Chapter 9].

Remark 2.2. Given $R > 0$, for $|z| < R$, consider

$$\wp(z) = \frac{1}{z^2} + \sum_{0<|w|<2R} + \sum_{|w|>2R} \left(\frac{1}{(z-w)^2} - \frac{1}{w^2}\right).$$

Then the second term is holomorphic and the first term exhibits poles of second degree at those lattice points in the disc $B(0, R)$. Therefore, $\wp$ has poles at L.

Next, we deduce an essential property between $\wp$ and $\wp'$.
To this end, let's revoke the identity

$$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k.$$

Thus, we have

$$\begin{aligned}
\wp(z) &= \frac{1}{z^2} + \sum_{w \in L^*} \left(\frac{1}{w^2}\left(\sum_{k=0}^{\infty}(k+1)\left(\frac{z}{w}\right)^k - \frac{1}{w^2}\right)\right) \\
&= \frac{1}{z^2} + \sum_{w \in L^*} \sum_{k=1}^{\infty} (k+1)\left(\frac{z^k}{w^{k+2}}\right) \\
&= \frac{1}{z^2} + \sum_{k=1}^{\infty} G_{k+2}(k+1)z^k \\
&= \frac{1}{z^2} + \sum_{k=1}^{\infty} G_{2k+2}(2k+1)z^{2k}.
\end{aligned}$$

Here, we denote by $G_k := \sum_{w \in L \setminus \{0\}} \frac{1}{w^k}$.

Proposition 2.3.

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3,$$

where $g_2 = 40G_4$ *and* $g_3 = 140G_6$.

Proof. Notice

$$\begin{aligned}
\wp &= \frac{1}{z^2} + 3G_4 z^2 + 5G_6 z^4 + \dots, \\
\wp^2 &= \frac{1}{z^4} + 6G_4 + 10G_6 z^2 + \dots, \\
\wp^3 &= \frac{1}{z^6} + \frac{9G_4}{z^2} + 15G_6 + \dots, \\
\wp' &= -\frac{2}{z^3} + 6G_4 z + 20G_6 z^3, \\
(\wp')^2 &= \frac{4}{z^6} - \frac{24G_4}{z^2} - 80G_6 + \dots.
\end{aligned}$$

Therefore, we find that

$$\begin{aligned}
(\wp')^2 - 4\wp^3 + 60G_4\wp &= \left(\frac{4-4}{z^6}\right) + \left(\frac{-24G_4 - 36G_4 + 60G_4}{z^2}\right) + (-80G_6 - 60G_6) + \dots \\
&= -140G_6 + \dots.
\end{aligned}$$

Since $(\wp')^2 - 4\wp^3 + 60G_4\wp$ is a holomorphic elliptic function, thus constant, it follows that

$$(\wp')^2 - 4\wp^3 + 60G_4\wp = -140G_6 = -g_3$$

by Liouville's theorem. □

3. Rational Parametrization and Group Structure

One can use the Weierstrass p-function to parametrize elliptic curves as follows:

$$\varphi : \mathbb{C}/L \to X(g_2(L), g_3(L)), \quad w \mapsto \begin{cases} [\wp(w) : \wp'(w) : 1], & w \notin L, \\ [0 : 1 : 0], & w \in L, \end{cases}$$

where $X(a, b)$ denotes the zero set of $F(X, Y, Z) = ZY^2 - 4X^3 + aXZ^2 + bZ^3$ in $\mathbb{P}^2$.

Fact: If $a^3 - 27b^2 \neq 0$, then there exists a lattice L with $g_2(L) = a$ and $g_3(L) = b$.

Proposition 3.1. *The map φ is an isomorphism between $\mathbb{C}/L$ and $X(g_2(L), g_3(L))$.*

Proof details can be adapted as needed. Here we focus on the lemmas and propositions formatting.

Lemma 3.2. *If $\wp(w_1) = \wp(w_2)$, then either $w_1 - w_2$ or $w_1 + w_2$ belongs to L.*

Proof. Suppose $w_1 - w_2 \notin L$. Consider the function $f(w) = \wp(w) - \wp(w_1) \in \mathcal{E}_L$. It has poles of second order at L. Therefore $f(w)$ has exactly two simple zeros at w_1, w_2 by Lemma 1.3. We can find $\alpha \in \mathbb{C}$ such that $\alpha + \Pi$ containing 0 and $\partial(\alpha + \Pi) \cap P(f) =$. We may choose $w_1', w_2' \in \alpha + \Pi$ where $w_1' - w_1, w_2' - w_2 \in L$. Revoking Lemma 1.4, we have that

$$\sum_{z \in \alpha + \Pi} ord_z(f) \cdot z = w_1' + w_2' - 2 \cdot 0 \in L$$

if $f \in \mathcal{E}_L$. That is $w_1 + w_2 \in L$. □

Lemma 3.3. *Let X be a compact Riemann surface and $f \in \mathcal{M}(X)$, the meromorphic function on X. If f is non-constant, then it is surjective.*

Proof. Confer [1][Theorem 2.7]. □

Proof of Proposition 3.1. **Injectivity:** Suppose $w_1, w_2 \in \mathbb{C}/L$ with $\varphi(w_1) = \varphi(w_2)$. If $\wp(w_1) = \wp(w_2)$, then either $w_1 - w_2$ or $w_1 + w_2 \in L$ by Lemma 3.2. Assume $w_1 + w_2 \in L$. Since $\wp'$ is odd, $\wp'(w_1) = -\wp'(-w_1) = -\wp'(w_2)$. If this term is non-zero, it contradicts $\varphi(w_1) = \varphi(w_2)$. If this term is zero, then $2w_1 \in L$, which implies $w_1 = w_2$.

Surjectivity: By Lemma 3.3, non-constant $\wp$ is surjective onto $\mathbb{P}^1$. Thus for any point $[x : y : 1] \in X(a, b) \cap U_2$, there exists a $w \in \mathbb{C}/L$ such that $\wp(w) = x$. By the formulas $(\wp')^2 = 4\wp^3 - a\wp - b$ and $y^2 = 4x^3 - ax - b$, it follows that either $(\wp(w), \wp'(w))$ or $(\wp(-w), \wp'(-w))$ equals (x, y), ensuring surjectivity.

Holomorphicity: For $w \notin L$, choose any neighborhood W excluding the origin with chart α, and let U_2 be the standard coordinate neighborhood with chart π_2. The composition $\pi_2 \circ \varphi \circ \alpha^{-1} = (\wp(\cdot), \wp'(\cdot))$ is holomorphic. Similarly, for $w = 0$, choose a neighborhood W without zeros of $\wp'$, with chart β and using neighborhood U_0, we find

$$\pi_0 \circ \varphi \circ \beta^{-1} = (\frac{\wp(\cdot)}{\wp'(\cdot)}, \frac{1}{\wp'(\cdot)})$$

is also holomorphic. This confirms the holomorphicity of φ. □

For the moment, we are trying to transfer the Abelian group structure from $(\mathbb{C}/L, +)$ to $X(a, b)$.

Lemma 3.4. *If $u, v, w \in \mathbb{C}$ with $u + v + w \in L$ and $\wp(u) \neq \wp(v)$, then*

$$\det \begin{pmatrix} 1 & \wp(u) & \wp'(u) \\ 1 & \wp(v) & \wp'(v) \\ 1 & \wp(w) & \wp'(w) \end{pmatrix} = 0.$$

Proof. Without loss of generality, We may assume $u, v, w \in L$.

If $\wp(u) \neq \wp(v)$, then there exists a unique pair $(a, b) \in \mathbb{C}^2$ such that

$$\begin{cases} b + a\wp'(u) = \wp(u), \\ b + a\wp'(v) = \wp(v). \end{cases} \tag{3.1}$$

On one hand, u, v are two distinct roots of the elliptic function $f = b + a\wp'(u) - \wp(u)$. This function, having poles of multiplicity three at L, must also have an extra simple zero.

Let us find $\alpha \in \mathbb{C}$ such that $\alpha + \Pi$ contains 0 and $\partial(\alpha + \Pi) \cap P(f) = \emptyset$. Choose $u', v' \in \alpha + \Pi$ where $u' - u, v' - v \in L$. Referring to the lemma above, we have $\sum_{z \in \alpha + \Pi} \mathrm{ord}_z(f) \cdot z = 0$ if $f \in \mathcal{E}_L$. If w' is the third zero of f, then $u' + v' + w' + 3 \cdot 0 = 0$. Hence, $w' = -(u' + v')$ and thus $w' - w \in L$. This results in the system

$$\begin{cases} b + a\wp'(u) = \wp(u), \\ b + a\wp'(v) = \wp(v), \\ b + a\wp'(-(u + v)) = \wp(-(u + v)), \end{cases} \tag{3.2}$$

being linearly dependent, or equivalently,

$$\det \begin{pmatrix} 1 & \wp(u) & \wp'(u) \\ 1 & \wp(v) & \wp'(v) \\ 1 & \wp(u + v) & -\wp'(u + v) \end{pmatrix} = 0.$$

□

To transition the group structure from $\mathbb{C}/L$ to $X(a,b)$, we define the operation $p+q = \varphi(\varphi^{-1}(p)+\varphi^{-1}(q))$ with the identity element $o := \varphi^{-1}(0) = [0:1:0]$. Notice, the inverse of $p = [\wp(w), \wp'(w):1]$ is given by $-p = [\wp(-w), \wp'(-w):1] = [\wp(w), -\wp'(w):1]$.

Remark 3.5. We reinterpret the lemma in a more geometric manner: if $p+q+r=o$, then they are collinear. This confirms the group structure on $X(a,b)$ as dispicted in Figure 3.1.

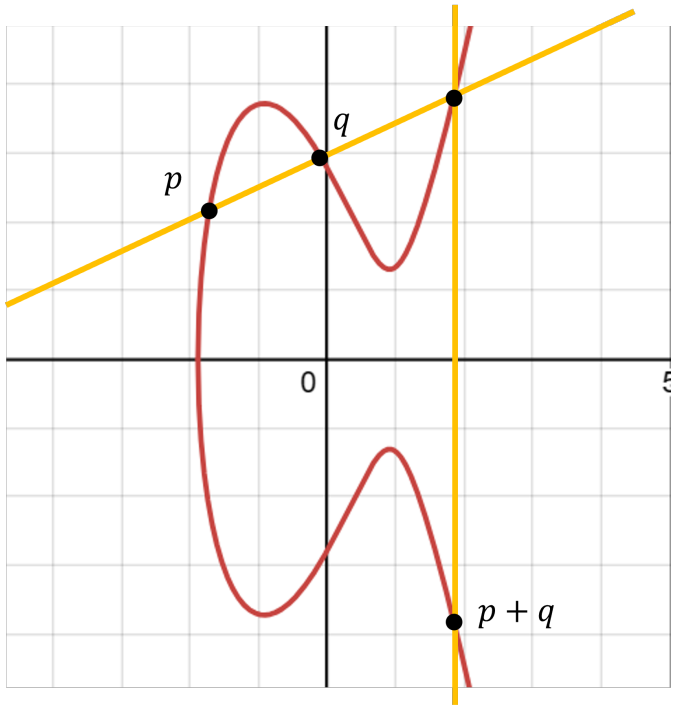

FIGURE 3.1. Group law on elliptic curve

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